

SHASIN MADUSHAN

Data Science Undergraduate | Aspiring Cloud, Data & ML Engineer

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PROFESSIONAL SUMMARY

Results-driven Data Science Undergraduate with academic and hands-on exposure building end-to-end data processing pipelines, optimizing complex SQL queries, and deploying business intelligence architectures. Proven ability to clean and validate high-volume structured and unstructured data, map relational environments, and build live visual reporting models via Power BI. Highly competent in Python-based predictive modeling and machine learning workflows, with a forward-looking roadmap specializing in cloud ecosystems like Databricks, Azure Data Factory, and scalable integrations. An analytical problem-solver eager to contribute to cutting-edge cloud and AI initiatives.

CORE COMPETENCIES & TECHNICAL SKILLS

Data Pipelines & Integration: ETL/ELT Architecture Design, SSIS Workflows, Real-Time Data Streaming, Ingestion Management, Pipeline Health Monitoring.

Cloud & Distributed Systems (Exposure): Databricks Frameworks, Azure Architecture Concepts, AWS Cloud Infrastructure, Confluent Cloud Kafka Ingestion.

Databases & Query Optimization: SQL (Advanced Query Optimization), Microsoft SQL Server, PostgreSQL, MySQL, MongoDB, Star Schema Design.

Data Analysis, BI & AI: Power BI Dashboard Development, Data Visualization, Trend & Pattern Discovery, Python (Pandas, Seaborn), Machine Learning (XGBoost, Random Forest), R Scripting, SQL Server Analysis Services (SSAS) Cubes.

Core Engineering & Software: Java, Spring Boot, Object-Oriented Programming (OOP), Node.js, JavaScript, Technical Documentation.

EDUCATION

BSc (Hons) in Information Technology, Specializing in Data Science

2023 – Present

Sri Lanka Institute of Information Technology (SLIIT)

- Academic Excellence:** Current GPA: 3.67 | Recipient of the Dean's List Award for Academic Excellence.
- Relational Database Distinction:** Earned the HackerRank SQL Gold Badge for advanced database query optimization, data validation, and multi-table joining.
- Continuous Technical Growth:** Actively completing KodeKloud's 50 Days of AWS Challenge (Day 28/50).

G.C.E. Advanced Level (A/L)

2022

St. Anne's College, Kurunegala, Sri Lanka

- Academic Profile:** Chemistry – B | Combined Mathematics – C | Physics – C (Z-Score: 1.0533)

TECHNICAL PIPELINES & ANALYTICAL PROJECTS

Olist E-Commerce Real-Time Data Streaming Pipeline

End-to-End Ingestion & Processing Architecture | Solo Project

Tech Stack: Python, Apache Kafka, AWS RDS (PostgreSQL), Cloud SQL, Confluent Cloud

- Data Processing & Streaming:** Engineered a custom Python/Kafka producer script to cleanly extract and stream over 300,000 raw transactional records into a cloud communication layer, successfully handling unstructured real-time event simulation.
- Pipeline Optimization:** Prevented API rate limits and data drops by programming a Kafka consumer engine that temporarily buffers live transactional streams in-memory to execute efficient bulk database insertions (5,000 records/batch).

- **Data Modeling & Visualization:** Designed a structured Bronze-to-Gold data layer by executing optimized SQL scripts to join, clean, and aggregate multi-source files into a single 99,000+ row reporting view; utilized Pandas and Seaborn to extract automated business trends.
- **Cloud Scale Roadmap:** Actively designing V2.0 migration to move data processing into a cloud environment utilizing a distributed PySpark/Databricks cluster and managing automated workflows via Apache Airflow DAG scheduling.

Global Airbnb Market Analysis

Data Engineering, NLP & Predictive Modeling

Stack: DuckDB, Python (VADER), R, Power BI

- **Data Architecture:** Built an ELT pipeline processing 2.1M+ global records (London, Tokyo, NYC) using DuckDB for high-speed columnar aggregation and harmonized complex multi-currency strings.
- **NLP & AI Sentiment:** Engineered a custom Python micro-batching script to safely process 2M+ unstructured text reviews through a VADER sentiment engine, prioritizing local hardware stability.

Predictive Analytics: Trained Random Forest regression models to map pricing predictability. Substantiated findings via Welch's t-tests ($p < 0.001$) and Cohen's d effect size

Banking Data Warehouse & Business Intelligence System

Data Integration, ETL Pipelines & BI Delivery | Academic Project

Tech Stack: Microsoft SQL Server, SSIS, SSAS, Power BI, Star Schema

- **ETL Pipeline Orchestration:** Built and maintained high-performance SSIS ETL pipelines to extract, clean, validate, and load high-volume banking transactional records from staging areas into a structured repository.
- **Data Modeling & Lineage:** Modeled a high-performance Star Schema architecture utilizing Slowly Changing Dimension (SCD Type 1 & Type 2) techniques to ensure historical data lineage, auditing, and complete data integrity.
- **Trend Analysis & Reporting:** Built SSAS OLAP multidimensional cubes and created interactive Power BI executive dashboards linked via live connections to discover business trends, seasonal pattern anomalies, and performance KPIs.

Rotten Tomatoes Predictive Analytics System

Data Preparation, Statistical Inference & AI/ML Modeling

Tech Stack: Python, R, XGBoost, Random Forest, Gradient Boosting, Data Wrangling

- **AI/ML Workflow & Modeling:** Executed a rigorous data science workflow to prepare data, run feature engineering, and train advanced machine learning predictive models (XGBoost, Random Forest) to evaluate and forecast movie rating trends.
- **Statistical Validation:** Conducted comprehensive exploratory analysis and non-parametric statistical tests (Shapiro-Wilk, Spearman correlation) to validate variable distributions and minimize model prediction errors (RMSE).

Smart Campus Operation Hub

Full-Stack Enterprise Logic

Tech Stack: Java, Spring Boot, React, MySQL

- **Validation Systems:** Developed a strict chronological conflict-validation script within the backend engine to remove data synchronization issues and schedule overlap errors completely.

CERTIFICATIONS & BADGES

- HackerRank SQL Gold Badge – Verified Advanced Database Query Specialist
- Event Streaming with Apache Kafka – KodeKloud
- Data Engineering Fundamentals – KodeKloud
- MongoDB Data Modeling Path – MongoDB University
- AWS Cloud Crash Course – KodeKloud
- Harvard University CS50P (Python)